Sana Iqbal Malik

AI Backend Engineer

Contact

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Skills

- Python, FastAPI, Pydantic
- LLM workflows, prompt design
- RAG, pgvector, PostgreSQL
- pytest, GitHub Actions
- Docker, AWS ECS

Education

MS Computer Science, NUST, 2020

BS Software Engineering, 2018

Experience

AI Backend Engineer, Meezan Claims Tech (2021 - present)

- Built a claims triage workflow that reads insurance emails in four steps: classify, extract fields, validate against policy data, route to an adjuster. It cut manual triage time by 62 percent.
- Wrote the FastAPI service behind it with Pydantic v2 validation, typed 422 and 409 errors with readable messages, and an OpenAPI contract used by two frontend teams.
- Built document question answering over 12,000 policy PDFs using pgvector search. Every answer shows the exact supporting passage with document id and page, and answers without a passage are refused.
- Designed the multi-tenant PostgreSQL schema with row level security on org_id. CI runs tenant isolation tests that try to read another customer's claims and must fail.
- Wrote 180 pytest tests, deployed with Docker and GitHub Actions to AWS ECS using blue-green releases, and added alerts and automatic rollback for failed health checks.

Software Engineer, Karachi Data Works (2018 - 2021)

- Built Python REST services for invoice processing and added retry and dead-letter handling for failed jobs.
- Introduced code review and automated testing, raising coverage from 20 to 78 percent.